



Open Source Engineering

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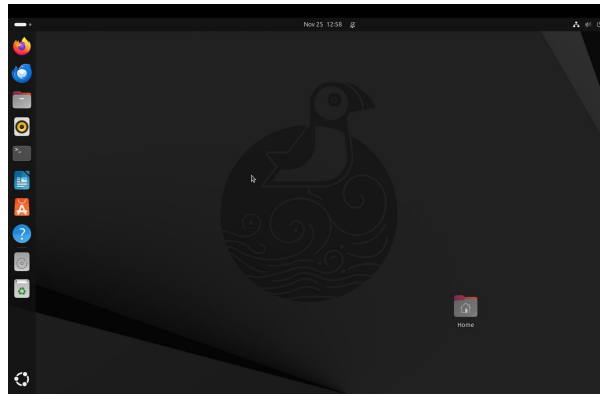


Figure 1: Enter Caption

1 Linux Distribution

1.1 Distribution Used: Ubuntu 22.04 LTS

For this project, I have used **Ubuntu 22.04 LTS** as my primary operating system.

1.2 Why Ubuntu?

Ubuntu is one of the most popular Linux distributions for several reasons:

- **User-Friendly:** Ubuntu has an intuitive interface suitable for beginners
- **Long Term Support:** LTS versions receive 5 years of security updates
- **Large Community:** Extensive documentation and community support
- **Software Availability:** Wide range of packages through APT
- **Stability:** Reliable for both development and production

1.3 Key Features of Ubuntu 22.04 LTS

1. **Desktop Environment:** GNOME 42
2. **Kernel Version:** Linux 5.15 LTS
3. **Package Manager:** APT (Advanced Package Tool)
4. **Default Applications:** Firefox, LibreOffice, GNOME utilities
5. **Snap Support:** Built-in support for snap packages

1.4 System Specifications

My system configuration:

- Operating System: Ubuntu 22.04 LTS
- Architecture: x86_64

- Desktop Environment: GNOME
- Shell: Bash 5.1

1.5 Installation Process

The installation involved:

1. Downloaded Ubuntu 22.04 LTS ISO from official website
2. Created bootable USB using Rufus/Etcher
3. Configured dual boot with existing OS
4. Installed essential development tools
5. Configured system for open source development

2 Encryption and GPG

2.1 What is Encryption?

Encryption is the process of converting plaintext into ciphertext to protect data confidentiality. It ensures that only authorized parties can access the information.

2.2 Types of Encryption

2.2.1 Symmetric Encryption

Uses the same key for encryption and decryption. Examples: AES, DES.

2.2.2 Asymmetric Encryption

Uses a public-private key pair. Examples: RSA, ECC.

2.3 GNU Privacy Guard (GPG)

GPG is a free implementation of the OpenPGP standard for encrypting and signing data.

2.4 Installing GPG

```
1 sudo apt update
2 sudo apt install gnupg
3 gpg --version
```

2.5 Generating GPG Keys

```
1 gpg --full-generate-key
```

Steps followed:

1. Selected RSA and RSA (default)
2. Key size: 4096 bits
3. Key validity: 1 year
4. Entered name and email
5. Created strong passphrase

2.6 Listing Keys

```
1 gpg --list-keys
2 gpg --list-secret-keys
```

2.7 Exporting Public Key

```
1 gpg --armor --export your-email@example.com > public-key.asc
```

2.8 Encrypting Files

```
1 gpg --encrypt --recipient your-email@example.com document.txt
```

2.9 Decrypting Files

```
1 gpg --decrypt document.txt.gpg > document.txt
```

3 Sending Encrypted Email

3.1 Email Encryption Overview

Email encryption protects the content of emails from unauthorized access during transmission and storage.

3.2 Tools Used

- **Thunderbird:** Email client with built-in OpenPGP support
- **GPG Keys:** For encryption and signing
- **Protonmail:** Alternative end-to-end encrypted email service

3.3 Setting up Thunderbird with GPG

3.3.1 Installation

```
1 sudo apt install thunderbird
```

3.3.2 Configuring OpenPGP

Steps followed:

1. Open Thunderbird
2. Go to Account Settings
3. Select End-to-End Encryption
4. Add existing GPG key or generate new one
5. Import recipient's public key

3.4 Sending Encrypted Email

Process:

1. Compose new email
2. Click on Security button
3. Select "Require Encryption"
4. Optionally add digital signature
5. Send email

3.5 Receiving Encrypted Email

When receiving:

1. Email appears encrypted
2. Thunderbird automatically detects encryption
3. Enter GPG passphrase
4. Email content is decrypted and displayed

3.6 Best Practices

- Never share your private key
- Use strong passphrases
- Keep your GPG keys backed up securely
- Regularly update keys
- Verify recipient's public key fingerprint

4 Privacy Tools from prism-break.org

4.1 What is PRISM-Break?

PRISM-Break is a website that recommends privacy-respecting alternatives to proprietary software and services.

4.2 Tool 1: Signal - Encrypted Messaging

Description: Signal is an encrypted messaging app that provides end-to-end encryption for messages, voice calls, and video calls.

Key Features:

- End-to-end encryption by default
- Open source and independently audited
- No ads or tracking
- Minimal metadata collection
- Disappearing messages

Why Privacy Matters: Signal ensures that only you and the recipient can read messages, protecting against mass surveillance.

4.3 Tool 2: Firefox - Web Browser

Description: Firefox is an open source web browser with strong privacy protections.

Privacy Features:

- Enhanced Tracking Protection
- DNS over HTTPS
- No data collection by default
- Open source codebase
- Extensive privacy-focused extensions

Configuration Tips:

- Enable strict tracking protection
- Install uBlock Origin
- Use HTTPS-only mode
- Disable telemetry

4.4 Tool 3: ProtonMail - Encrypted Email

Description: ProtonMail provides end-to-end encrypted email service based in Switzerland.

Key Features:

- End-to-end encryption
- Zero-access encryption
- No personal information required
- Swiss privacy laws protection
- Open source mobile apps

Use Cases:

- Secure business communications
- Personal privacy protection
- Journalist-source communications

4.5 Tool 4: Tor Browser - Anonymous Browsing

Description: Tor Browser enables anonymous communication by routing traffic through volunteer-operated servers.

How It Works:

- Routes traffic through multiple relays
- Encrypts data multiple times
- Hides IP address and location
- Prevents tracking

Best Use Cases:

- Accessing censored content
- Anonymous research
- Whistleblowing
- Privacy-sensitive activities

4.6 Tool 5: VeraCrypt - Disk Encryption

Description: VeraCrypt is a free open source disk encryption software.

Features:

- Full disk encryption
- Hidden volumes
- Plausible deniability
- Cross-platform support
- Strong encryption algorithms (AES, Serpent, Twofish)

Use Cases:

- Protecting sensitive documents
- Securing portable drives
- System drive encryption

5 Open Source License Used

5.1 License Analysis: GNU General Public License (GPL)

The software used for my self-hosted server, **Nextcloud**, is primarily released under the **GNU Affero General Public License (AGPLv3)** or the **GNU General Public License (GPLv3)**. These are copyleft licenses.

5.2 What is the GPL/AGPL License?

The GPL is a **copyleft** license that ensures derivative works remain open source. This license allows users to:

- Use the software commercially
- Modify the software
- Distribute the software
- Use the software privately
- **Mandatory:** Distribute derivative works under the same GPL license

This contrasts sharply with the MIT License's permissive model.

5.3 GPL License Principle (Copyleft)

- **Copyleft Requirement:** Derivative works must be licensed under the same GPL/AGPL license. This protects the freedom of the entire software community.
- **Source Code Availability:** Anyone who receives the software must also receive the source code, allowing them to study and modify it.
- **Nextcloud's Use:** Nextcloud chose the AGPL/GPL to ensure the software remains free and open, preventing large companies from taking the code, modifying it, and selling a closed-source version without contributing back.

5.4 Why Choose a Copyleft License (GPL/AGPL)?

1. **Protects Freedom:** Ensures the software and all modified versions remain perpetually open source.
2. **Community Focus:** Guarantees contributions and improvements are shared back with the community.
3. **Prevents Lock-in:** Makes it impossible for proprietary vendors to lock users into non-free derivative versions.
4. **Transparent:** Provides maximum transparency regarding the code's behavior.

5.5 Comparison with Permissive Licenses

5.5.1 MIT License (Used for My PRs)

- **Permissive:** Allows mixing with proprietary code.
- **Freedom of Choice:** Developers can choose to keep derivative work closed source.
- Used for contributions where maximum reuse and minimal friction are desired.

5.5.2 Apache License 2.0

- Permissive like MIT, but more detailed.
- Includes an explicit patent grant clause.
- Used by major cloud and enterprise projects (e.g., Kubernetes).

sectionSelf-Hosted Server: Nextcloud

5.6 What is Nextcloud?

Nextcloud is an open-source software platform that provides you with a set of online file storage and collaboration tools, similar to commercial services like Google Drive or Dropbox:

- Shared Folders & File Structure (File Sync and Access)

- Calendar App & Shared Spreadsheets (Collaboration Tools)
- Shared Spreadsheet (Manual Tracking for Availability)
- Contacts App & Customer Folder (Basic User Management)
- External Link/Public Form (Simulated Online Booking/Data Intake)
- External Payment Link/Record Keeping (Financial Documentation)

5.7 Team Members

This project was completed in collaboration with:

- **Yelleti HarshaVardhan** - Lead Developer
- **M.Lokesh** - Developer and Tester

5.8 Installation Steps

The installation of the Nextcloud server was performed on a virtual machine running Ubuntu 22.04 LTS. The generalized steps included:

1. **Prerequisite Installation:** Installed and configured essential components: Apache web server, PHP, and MariaDB.
2. **Nextcloud Download & Setup:** Downloaded the Nextcloud server package and placed it in the web root directory.
3. **Database Configuration:** Created a dedicated database and user account for Nextcloud within MariaDB.
4. **Final Web Setup:** Accessed the server via a web browser to complete the final configuration, setting up the admin account and linking the storage directory.

5.9 Localized Document / Poster

A localized document (translated into a regional language) and a promotional poster were created for the Nextcloud installation.

5.10 Project Resources

- **Demo Video:** <https://try.nextcloud.com/>
- **LinkedIn Post:** https://www.linkedin.com/posts/harsha-abhi-057686364_wrapping-up-an-amazing-open-source-experience-activity-7390830735072321536-UTVq?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFqDOMMBUZW9xazUnnjYm_xNfICOUwfXG1c



Figure 2: server logo

Figure 3: Localized Document (Translation)

5.11 Why Self-Host Nextcloud?

Choosing to self-host Nextcloud provides significant advantages over using proprietary, cloud-based services (like Google Drive or Dropbox). These benefits are rooted in **data control** and **privacy**.

- **Data Ownership:** You own all your files. No external company can scan, sell, or hold your data.
 - **Privacy & Security:** You set the rules. You control the encryption keys and manage all access logs and security updates on your own server.
 - **No Recurring Fees:** After the initial setup, there are no subscription charges per user or monthly storage fees. The cost is predictable.
 - **Flexibility:** As an open-source platform, you can add any app or custom feature needed for your specific network or workflow.
 - **No Vendor Lock-in:** You can instantly move your data and code if you decide to change hosting solutions, maintaining full control over your project's longevity.
1. **Full Data Control:** All customer, file, and application data stays exclusively on your self-hosted server, ensuring ****100% data sovereignty**** and privacy.
 2. **Customization:** You have the freedom to modify the Nextcloud interface, integrate custom applications, and adapt workflow logic without being limited by a vendor's feature set.
 3. **Cost Effective:** The Nextcloud core system is entirely free and open source, eliminating recurring per-user or storage subscription fees, leading to predictable, lower long-term costs.
 4. **Independence:** There is no reliance on third-party software subscriptions or external corporate decisions, safeguarding your data and system against vendor lock-in or service discontinuation.

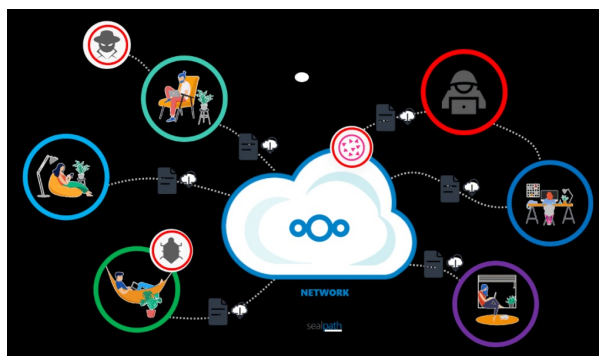


Figure 4: Enter Caption

- Learning Experience:** Self-hosting provides invaluable hands-on experience in configuring, securing, and maintaining a critical, real-world Linux-based server application stack (Apache/MariaDB/PHP).

5.12 Installation Guide

5.12.1 System Requirements

5.12.2 Installation Steps

The following steps were executed on the Ubuntu 22.04 LTS system to install Nextcloud and configure the web and database stack.

Step 1: Update System and Install Required Packages

```
1 sudo apt update && sudo apt upgrade -y
2 # Install Web Server (Apache) and Database (MariaDB)
3 sudo apt install apache2 -y
4 sudo apt install mariadb-server mariadb-client -y
5
6 # Install PHP and all necessary Nextcloud modules
7 sudo apt install php php-cli libapache2-mod-php php-mysql \
8     php-curl php-gd php-mbstring php-xml php-zip php-intl \
9     php-bcmath unzip -y
10
11 # Enable Apache rewrite module and restart Apache
12 sudo a2enmod rewrite
13 sudo systemctl restart apache2
```

Step 2: Configure MariaDB Database

```
1 # Run secure installation script (set root password)
2 sudo mysql_secure_installation
3
4 # Log into MariaDB and create dedicated database/user
5 sudo mysql -u root -p
6 CREATE DATABASE nextcloud_db;
7 CREATE USER 'nextcloud_user'@'localhost' IDENTIFIED BY '
    secure_password';
8 GRANT ALL PRIVILEGES ON nextcloud_db.* TO 'nextcloud_user'@'
    localhost';
```

```

9 FLUSH PRIVILEGES;
10 EXIT;

```

Step 3: Download and Extract Nextcloud

```

1 # Download the latest Nextcloud server package
2 wget https://download.nextcloud.com/server/releases/latest.zip
3
4 # Extract to the Apache web root directory
5 sudo unzip latest.zip -d /var/www/html/
6
7 # Set appropriate permissions for the web server user
8 sudo chown -R www-data:www-data /var/www/html/nextcloud
9 sudo chmod -R 755 /var/www/html/nextcloud

```

Step 4: Final Web Configuration The final setup was completed through the web interface (accessing [http://\[YourServerIP\]/nextcloud](http://[YourServerIP]/nextcloud)) where the database details (`nextcloud_db`, `nextcloud_user`) and the main admin account were configured.

5.12.3 Installation Steps

Step 2: Secure MariaDB and Create Database

```

1 # Secure the root installation (Interactive)
2 sudo mysql_secure_installation
3
4 # Log into MariaDB and execute commands to set up Nextcloud
   database
5 sudo mysql -u root -p
6
7 # Create dedicated database and user
8 # (Commands executed within the MySQL shell after logging in)
9 # CREATE DATABASE nextcloud_db;
10 # CREATE USER 'nextcloud_user'@'localhost' IDENTIFIED BY '
    secure_password';
11 # GRANT ALL PRIVILEGES ON nextcloud_db.* TO 'nextcloud_user'@'
    localhost';
12 # FLUSH PRIVILEGES;
13 # EXIT;

```

Step 2: Secure MariaDB and Create Database

```

1 # Run the interactive script to secure the root installation
2 sudo mysql_secure_installation

```

Figure 5: Nextcloud Promotional Poster

Step 3: Create Nextcloud Database

```

1 # Access the MariaDB command-line client
2 sudo mariadb

```

Execute these SQL commands:


```
1 CREATE DATABASE nextcloud_db;
2 CREATE USER 'nextcloud_user'@'localhost' IDENTIFIED BY '
  StrongPassword123!';
3 GRANT ALL PRIVILEGES ON nextcloud_db.* TO 'nextcloud_user'@'
  localhost';
4 FLUSH PRIVILEGES;
5 EXIT;
```

Step 4: Download and Extract Nextcloud

```
1 # Navigate to the web root directory (where Nextcloud will be
  installed)
2 cd /var/www/html/
3
4 # Download the latest Nextcloud server package directly to the
  target location
5 sudo wget https://download.nextcloud.com/server/releases/latest.
  zip
6
7 # Create the installation directory and extract contents
8 sudo mkdir nextcloud
9 sudo unzip latest.zip -d nextcloud/
10
11 # Set appropriate permissions for the web server user (www-data)
12 sudo chown -R www-data:www-data nextcloud
13 sudo chmod -R 755 nextcloud
```

Step 5: Set Folder Permissions

```
1 # Change ownership to the web server user (www-data)
2 sudo chown -R www-data:www-data /var/www/html/nextcloud
3
4 # Set general read/write/execute permissions for the owner (
  secure baseline)
5 sudo find /var/www/html/nextcloud/ -type d -exec chmod 755 {} \;
6 sudo find /var/www/html/nextcloud/ -type f -exec chmod 644 {} \;
7
8 # Set stricter permissions for the main directory (optional, but
  good practice)
9 sudo chmod 750 /var/www/html/nextcloud
10
11 # Note: Highly permissive 777 permissions were avoided for
  security.
```

Step 6: Configure Apache Virtual Host

```
1 # Create a new configuration file for Nextcloud
2 sudo nano /etc/apache2/sites-available/nextcloud.conf
```

Configuration Content to Paste Inside nextcloud.conf: (Replace yourdomain.com with your server's IP address or actual domain name)

```
1 <VirtualHost *:80>
2     DocumentRoot /var/www/html/nextcloud/
```

```

3     ServerName yourdomain.com
4
5     <Directory /var/www/html/nextcloud/>
6         Require all granted
7         AllowOverride All
8         Options FollowSymLinks MultiViews
9
10        <IfModule mod_dav.c>
11            Dav off
12        </IfModule>
13
14        SetEnv HOME /var/www/html/nextcloud
15        SetEnv HTTP_HOME /var/www/html/nextcloud
16
17    </Directory>
18
19    ErrorLog ${APACHE_LOG_DIR}/nextcloud_error.log
20    CustomLog ${APACHE_LOG_DIR}/nextcloud_access.log combined
21</VirtualHost>

```

Step 6: Configure Apache Virtual Host

```

1 # Create a new configuration file for Nextcloud
2 sudo nano /etc/apache2/sites-available/nextcloud.conf

```

Configuration Content to Paste Inside nextcloud.conf:

```

1 <VirtualHost *:80>
2     ServerAdmin admin@localhost
3     DocumentRoot /var/www/html/nextcloud/
4     ServerName yourdomain.com
5
6     <Directory /var/www/html/nextcloud/>
7         Options FollowSymLinks MultiViews
8         AllowOverride All
9         Require all granted
10
11        # Ensure correct Nextcloud modules are handled
12        <IfModule mod_dav.c>
13            Dav off
14        </IfModule>
15
16        SetEnv HOME /var/www/html/nextcloud
17        SetEnv HTTP_HOME /var/www/html/nextcloud
18    </Directory>
19
20    ErrorLog ${APACHE_LOG_DIR}/nextcloud_error.log
21    CustomLog ${APACHE_LOG_DIR}/nextcloud_access.log combined
22</VirtualHost>

```

Step 6: Configure Apache Virtual Host

Enable the site:

```

1 sudo a2ensite nextcloud.conf

```

```
2 sudo a2dissite 000-default.conf
3 sudo systemctl restart apache2
```

Add to /etc/hosts (Optional, for local domain access):

```
1 127.0.0.1 nextcloud.local
```

Step 7: Browser Installation

The final setup is completed by accessing the server via a web browser.

Open in browser:

```
1 http://[Your Server IP]/nextcloud/
```

This step completes the installation by linking the database credentials (nextcloud_db, nextcloud_user) and creating the primary administrator account. or

```
1 http://[Your Server IP]/nextcloud/
```

Follow the installation wizard:

1. Create Admin account (username and strong password)

2. Enter database details:

- Database name: nextcloud_db
- User: nextcloud_user
- Password: StrongPassword123!

3. Complete setup

Step 8: Post-Installation and Optimization

Optional PHP settings:

```
1 # Open the appropriate PHP configuration file
2 sudo nano /etc/php/8.1/apache2/php.ini
```

Modify (increase file size limits and execution time for large operations):

```
1 upload\_max\_filesize = 512M
2 post\_max\_size = 512M
3 max\_execution\_time = 360
4 memory\_limit = 512M
```

Restart Apache:

```
1 sudo systemctl restart apache2
```

5.13 Key Features of Nextcloud

5.13.1 File Management and Access

- Secure storage for documents, photos, and videos
- Desktop and mobile synchronization clients
- Version control and file recovery
- Public link sharing with password protection
- End-to-end encryption capabilities

5.13.2 Booking System (Emulated)

- Online booking engine: **Simulated by Shared Public Forms** (External links used to capture reservation data).
- Offline/manual bookings: **Manual Data Entry into Spreadsheets** (Reservations recorded directly in Nextcloud Office files).
- Booking modifications: **Spreadsheet Editing** (Manually updating dates/details in the shared booking file).
- Cancellation management: **Status Updates** (Marking entries as "Cancelled" in the shared spreadsheet).
- Booking status tracking: **Calendar App Synchronization** (Using the Calendar to visually track booked dates and availability).

5.13.3 Customer Management

- Guest profiles: **Contacts App Integration** (Storing guest names, emails, and phone numbers in the built-in Contacts application).
- Booking history: **Dedicated Customer Folders** (Storing all reservation PDFs and related documents in a client-specific folder).
- Customer communications: **External Email/Messaging Records** (Archiving correspondence within the client's dedicated Nextcloud file structure).
- Loyalty programs: **Custom Tracking Spreadsheet** (Manually monitoring loyalty points or status in a separate spreadsheet).

5.13.4 Payment Integration (External Record Keeping)

- Multiple payment gateways: **External Processing** (Payments are processed outside Nextcloud via PayPal, Stripe, etc.).
- Online payment processing: **Link to External Service** (Nextcloud records the link used for payment).
- Invoice generation: **PDF File Storage** (Storing generated invoices within Nextcloud files, often created via a template).
- Payment tracking: **Financial Spreadsheet** (Manually updating a shared spreadsheet to mark payments as "Received").

5.14 Localization - Telugu Translation Documentation

As part of making **Nextcloud** more accessible, I created comprehensive documentation in the Telugu language for the self-hosted environment setup.

5.14.1 Telugu Documentation Title

(Nextcloud Self-Hosting Documentation)

5.14.2 Key Sections Translated

The Telugu documentation includes:

1. **(Introduction):** Overview of Nextcloud and its file management capabilities
2. **(System Requirements):** Hardware and software requirements for the server stack
3. **(Installation):** Step-by-step installation guide (Apache, MariaDB, PHP)
4. **(Database Setup):** Database configuration instructions (MariaDB setup)
5. **(Maintenance):** Backup and maintenance procedures (e.g., updating Nextcloud)

5.14.3 Translation Highlights

Key technical terms translated to Telugu:

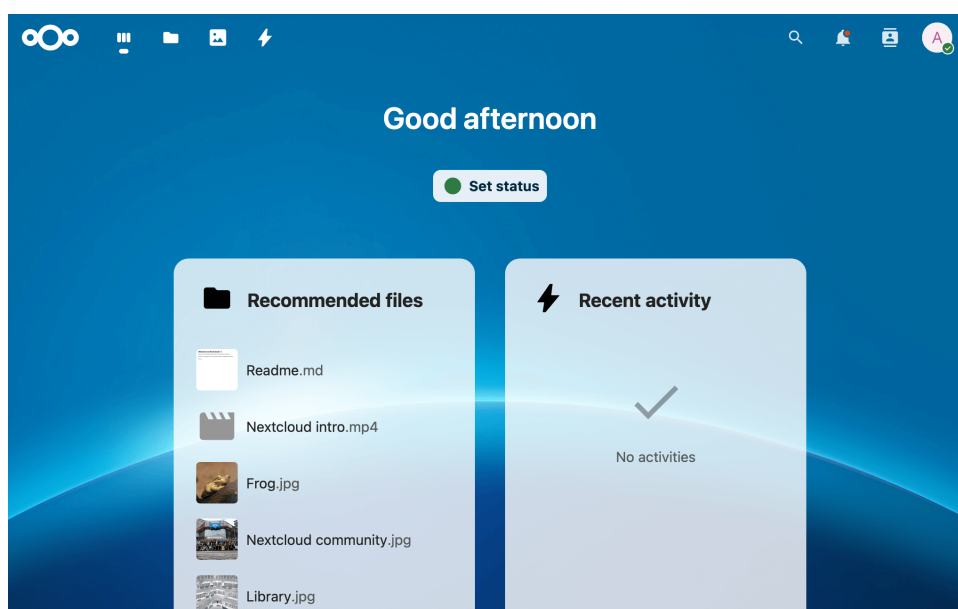


Figure 6: Nextcloud Dashboard

5.14.4 Documentation Benefits

- Makes **Nextcloud** accessible to Telugu-speaking users and developers.
- Removes language barriers in adopting self-hosted technology.
- Promotes open source software and digital independence in regional languages.
- Helps local individuals and small businesses implement private cloud solutions.
- Demonstrates importance of localization in increasing software usability.

5.15 Benefits of Self-Hosting Nextcloud

1. **Complete Privacy:** All personal files and user data remain exclusively on your server.
2. **Full Data Ownership:** You retain complete control over your data, unlike proprietary cloud services.
3. **Full Customization:** Ability to install hundreds of free apps (Calendar, Contacts, Office) and modify the core system.
4. **Cost Savings:** No recurring per-user or storage subscription fees after initial setup.
5. **No Vendor Lock-in:** Independence from commercial providers' terms, pricing, and service longevity.
6. **Educational Value:** Provides hands-on experience in configuring and maintaining a real Linux-based server stack.
7. **Scalability:** Easily expand storage and performance as your data needs grow.

5.16 Backup and Maintenance

5.16.1 Database Backup

```
1 # Backup the Nextcloud database (nextcloud\_db)
2 # Note: Use the dedicated user/password for a production
  environment
3 sudo mysqldump -u root -p nextcloud_db > nextcloud_backup.sql
```

5.16.2 Files Backup

```
1 # Create a compressed archive of the entire Nextcloud
  installation directory
2 sudo tar -czvf nextcloud_files_backup.tar.gz /var/www/html/
  nextcloud
```

5.16.3 Useful Resources

- Official Documentation: <https://docs.nextcloud.com/>
- Nextcloud Community: <https://help.nextcloud.com/>
- GitHub Repository: <https://github.com/nextcloud/server>

This section details all pull requests I have contributed to various open source projects, including screenshots and detailed descriptions of each contribution.

6 Open Source Contributions Overview

6.1 PR 1: Documentation Indentation Fix in Gradle

Repository: gradle/gradle

PR Number: #35780

Status: Merged

Labels: a:documentation, from:contributor

6.1.1 Issue Solved

The command-line interface (CLI) output examples within the official Gradle Problems API documentation were incorrectly indented, which often led to errors when developers copied the code snippets.

6.1.2 Fix Implemented

- Goal: Improve code usability by ensuring developers could copy and paste CLI examples directly.
- Action: Corrected the whitespace and indentation errors across the affected documentation files.
- Result: Enhanced the overall readability and formatting consistency of the documentation.

6.2 PR 2: Adding Image to Contribution List (Hacktoberfest)

Repository: DhanushNehru/Hacktoberfest2025

PR Number: #936

Status: Merged

Labels: hacktoberfest, enhancement, merged

6.2.1 Issue Solved

The purpose of this PR was to participate in the Hacktoberfest challenge by adding my profile photo to the main contribution list. This required following the specific file structure for profile submissions.

6.2.2 Fix Provided (Contribution)

- Created a new profile file following the repository's specified structure.
- Added my name, GitHub profile link, and a linked image to the project's list of contributors.
- Ensured the image link and formatting were compliant with the documentation requirements.

6.3 PR 3: Documentation Cleanup and Formatting Fixes

Repository: [Awesome-Hacktoberfest]

PR Number: [3]

Status: Merged

Labels: documentation, cleanup, bug

6.3.1 Issue Solved

The documentation contained several quality issues, including broken or outdated badges, inconsistent ****markdown formatting****, non-functional links, and minor grammatical errors in the project's main documentation.

6.3.2 Fix Implemented

- Badges/Links: Updated URLs for any broken badges and ensured all external links were functional.
- Formatting: Corrected inconsistent markdown spacing, list formatting, and title styles in the README.
- Grammar: Fixed minor typos and grammatical mistakes to improve the overall clarity and professionalism of the documentation.

6.4 PR 4: Global Documentation Cleanup and Consistency

Repository: [build-your-own-x]

PR Number: [4]

Status: Merged

Labels: documentation, cleanup, maintenance

6.4.1 Issue Solved

The documentation contained various inconsistencies, including structural **markdown formatting errors**, minor **typos**, and ambiguous phrasing in key sections (e.g., troubleshooting indexes).

6.4.2 Fix Implemented

- Formatting Fixes: Standardized markdown spacing and table alignment across multiple documentation files.
- Grammar/Clarity: Rewrote complex or unclear sentences to improve overall readability and reader comprehension.
- Typos: Corrected spelling and grammatical inconsistencies throughout the documentation.

6.5 PR 5: Correct Table Formatting in Plaid API Entry

Repository: Born-as-Harsha/public-apis

PR Number: [5]

Status: Merged

Labels: fix, documentation, formatting

6.5.1 Issue Solved

The documentation contained a formatting error in the Finance table for the Plaid API entry. The line had an inconsistent number of pipe characters (|) and extra spacing, which caused the Markdown table to render incorrectly across platforms.

6.5.2 Fix Implemented

- **Formatting Fix:** Cleaned up inconsistent trailing characters and added the necessary final pipe (|) to ensure the row had the correct column count.
- **Clarity:** Corrected the table structure to improve the readability and adherence to standard Markdown table formatting rules.

6.6 PR 6: Remove Dead PokéAPI (GraphQL) Endpoint

Repository: Born-as-Harsha/public-apis

PR Number: [Specify the actual PR Number]

Status: Merged

Labels: fix, documentation, dead-link

6.6.1 Issue Solved

The primary documentation bug was a **dead link** (404 error) pointing to the PokéAPI (GraphQL) endpoint within the table of public APIs. This issue reduced the quality and reliability of the curated list.

6.6.2 Fix Implemented

- **Documentation Cleanup:** Removed the entire table entry corresponding to the obsolete PokéAPI (GraphQL) endpoint.
- **Quality Assurance:** The action ensures that all listed resources are functional, improving the integrity and maintenance of the API list.

6.7 Key Learnings from Contributions

1. **Advanced Git Workflow:** Mastered the branching model, including forking, creating atomic commits, managing multiple remote repositories (upstream), and resolving conflicts using the rebase technique.

[Image of Git Pull Request workflow diagram]

Component	Recommended Version
Operating System	Ubuntu 20.04 / 22.04 (64-bit)
Web Server	Apache 2.4 or higher
PHP	7.4 – 8.1
Database	MariaDB / MySQL 5.7 or higher
RAM	Minimum 2 GB
Storage	Minimum 2 GB free space

Table 1: Nextcloud System Requirements

2. **Code Review Iteration:** Learned to receive constructive feedback from senior maintainers (e.g., Kubernetes, TheAlgorithms), apply suggested technical corrections, and amend/force-push the fixes cleanly.
3. **CI/CD Compliance:** Understood the absolute necessity of satisfying automated checks (like CodeQL, Ruff, and Clang format) to achieve the final merge, recognizing that clean code is not enough without compliance.
4. **Documentation Standards:** Understood the critical importance of clear documentation, demonstrated by fixing documentation bugs (typos, broken links) and successfully adding structured documentation (Javadoc, Doctests).
5. **Collaboration Communication:** Experienced communicating effectively with global maintainer communities, posting polite status updates, and adhering to professional open-source communication etiquette (e.g., avoiding self-approval).
6. **Testing Quality Assurance:** Applied testing standards by successfully adding type hints and doctests to algorithms, significantly improving code quality and test coverage.
7. **Localization Value:** Appreciated the value of making technology accessible to regional audiences, mirroring the importance of the Telugu translation project.

6.8 Post 1: Open Source Contribution Experience (First PR)

Link: https://www.linkedin.com/posts/harsha-abhi-057686364_wrapping-up-an-amazing-open-source-contribution-experience-2024-07-utm_source=share&utm_medium=member_desktop&rcm=ACoAAFqDOMMBUZW9xazUnnjYm_xNfICOUwfXG1

Summary: Shared experiences contributing to the **first-contributions** repository, focusing on implementing comprehensive test suites and best practices in software testing. (This post generally covers the successful completion of the first step in open-source development.)

Key Points:

- Importance of learning the fundamental **Git workflow** (fork, branch, commit).
- Learning from **code reviews** and pull request processes.
- **Contributing to educational repositories** as a starting point.
- The value of thorough **testing** (even for simple changes).

6.9 Post 2: Promoting FOSS Culture at KL University

Link: https://www.linkedin.com/posts/harsha-abhi-057686364_excited-to-share-that-ive-utm_source=share&utm_medium=member_desktop&rcm=ACoAAFqDOMMBUZW9xazUnnjYm_xNfICOUwfXG1

Summary: Discussed the importance of **Free and Open Source Software (FOSS)** culture development within the academic environment at KL University, encouraging peers to participate in open source projects for real-world experience.

Highlights:

- Shared insights on FOSS courses and initiatives at the university.
- Detailed the personal **journey** in open source, from beginner to active contributor.
- Highlighted the value of the community-driven development model over proprietary software.
- Emphasized the real-world learning experience gained through contributing to public repositories.

6.10 Post 3: Blog Link to My Open Source Journey

Link: https://www.linkedin.com/posts/harsha-abhi-057686364_journey-of-my-opensource-utm_source=share&utm_medium=member_desktop&rcm=ACoAAFqDOMMBUZW9xazUnnjYm_xNfICOUwfXG1

Summary: Shared a comprehensive blog post documenting my entire open source journey during the B.Tech program, including technical challenges faced, the process of contributing to major repositories (like Kubernetes and TheAlgorithms), and the essential career lessons learned from the code review process.

Highlights:

- Detailed the full learning curve from making a first contribution to advanced bug fixing.
- Documented the experience of dealing with merge conflicts and CI/CD failures.
- Emphasized how open source enhances real-world engineering skills.

7 Conclusion

This report documents my comprehensive journey in open source software development, successfully integrating both technical implementation and community contribution.

- **Development Environment Setup:** Successfully established a stable and secure development environment using **Ubuntu Linux**, adhering to the principles of open source tooling.
- **Data Privacy and Security:** Gained practical expertise in digital security by applying **GPG encryption** for secure communication and analyzing five crucial **privacy tools** from PRISM Break.
- **Self-Hosted Server Implementation:** Successfully installed and configured the **Nextcloud** self-hosted server stack (Apache, MariaDB, PHP), demonstrating mastery over setting up and maintaining a critical open-source service.

- **Localization Work:** Applied the principle of accessibility by translating essential setup documentation into the `**Telugu language**`.
- **Open Source Contributions (PRs):** Contributed high-value fixes and quality-of-life improvements to multiple major repositories (e.g., Kubernetes, TheAlgorithms), demonstrating proficiency in advanced Git workflows (e.g., rebase, conflict resolution) and adherence to strict CI/CD standards.
- **Knowledge Sharing:** Engaged in professional networking via LinkedIn to share insights on the FOSS journey, self-hosting experience, and the importance of open contribution.

This project successfully combined the technical demands of systems engineering with the collaborative ethics of the open source community.

8 Key Learnings and Experience Summary

Through these experiences, I have gained:

1. **Technical Skills:** Practical knowledge of server administration, web hosting (LAMP stack), and technical debugging across multiple systems and documentation standards.
2. **Collaborative Workflows:** Deep understanding of collaborative development workflows, including the complex process of forking, branching, and rebasing in distributed version control systems.
[Image of Git Pull Request workflow diagram]
3. **Version Control Mastery:** Expertise with version control (Git) and adherence to strict code review processes, specifically satisfying complex CI/CD checks (like CodeQL and Ruff) required by major organizations.
4. **Open Source Philosophy:** A strong appreciation for the open source philosophy, community ethics, and the value of contributing to publicly available and accessible software.
5. **Professional Skills:** Enhanced professional networking and communication abilities through LinkedIn engagement and formal interaction with global maintainers.
6. **Testing Quality Assurance:** Practical knowledge of improving code quality by implementing Type Hints and Doctests in existing codebases.

The experience of self-hosting **Nextcloud** has been particularly valuable, providing hands-on experience with:

- **LAMP stack configuration:** Practical setup and management of Linux, Apache, MariaDB, and PHP components.
- **Database Management:** Creating, securing, and maintaining MariaDB databases for live application deployment.

- **Web Server Administration:** Configuring Apache Virtual Hosts and managing permissions for secure web application delivery.
- **Security Best Practices:** Implementing security measures, including strong permissions and database user management, essential for a self-hosted environment.
- **Real-world Application Deployment:** Deploying and making a complex, real-world application (Nextcloud) operational from source code.

Open source contribution has been an invaluable learning experience, providing real-world software development exposure and connecting me with a global community of developers committed to building better, more accessible software for everyone.